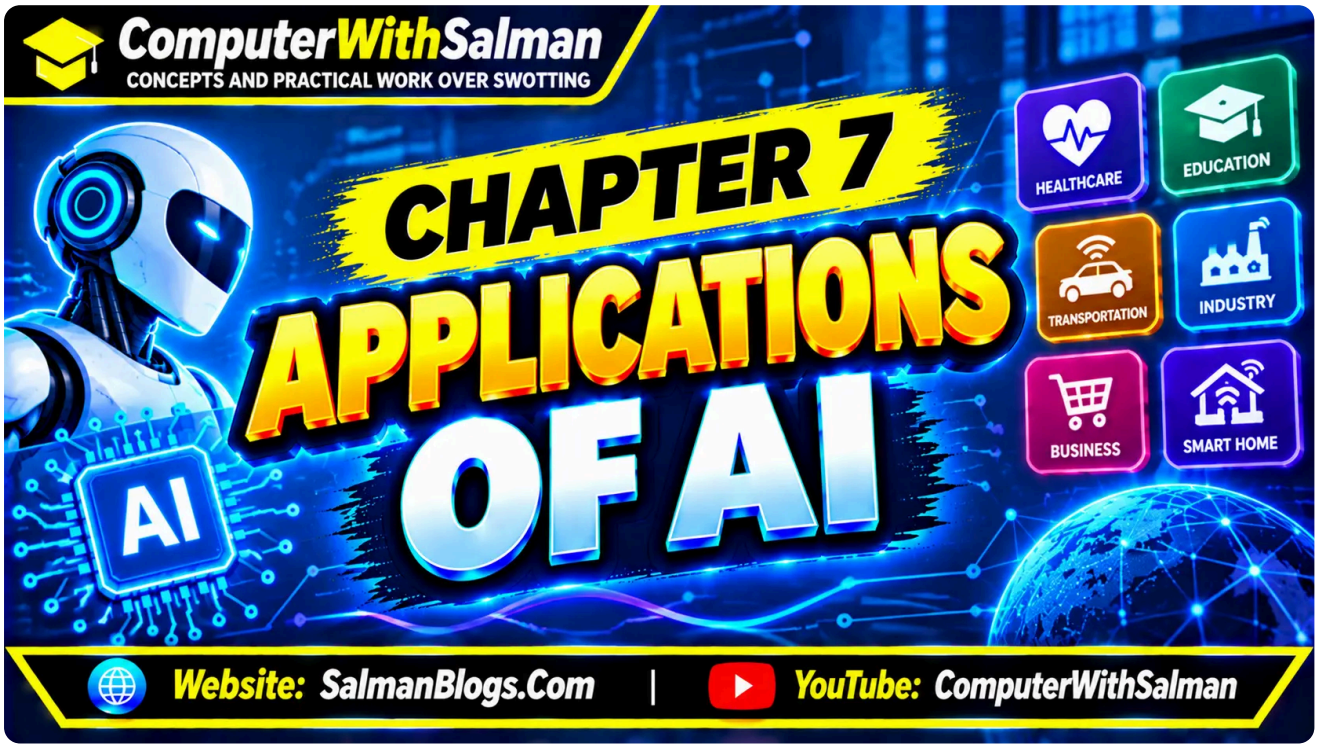
# Class 10 Computer Science Chapter 7: Applications of Artificial Intelligence (AI)

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These lecture notes are prepared by **Salman Ahmad**, who is both a **teacher** and a **web developer**. The purpose of this material is to make the topic clear and useful for students of Class 10 Computer Science.

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## Artificial Intelligence (AI)

Artificial Intelligence (AI) is a technology that enables computers and machines to perform tasks that normally require human intelligence, such as understanding language, recognizing images, and solving problems.

### Examples of AI [Mention any 1 or 2 Examples during writing exam]

- **Google Assistant:** Understands voice commands and responds to users.
- **ChatGPT:** Answers questions, explains topics, and helps users write content.
- **Self-Driving Cars:** Use AI to recognize roads, traffic signs, and other vehicles.
- **YouTube:** Recommends videos based on users' interests.
- **Face Recognition:** Identifies people by analyzing their facial features.
- **Online Chatbots:** Answer customers' questions on websites.
- **Medical Diagnosis:** Helps doctors identify diseases by analyzing medical images and data.
- **Robots:** Use AI to perform tasks such as sorting objects and working in factories.

## Impact of AI on Jobs and Work

- **Automation:** AI-powered machines and robots can perform tasks that were previously done by humans.
- **Faster Work:** AI helps people complete tasks faster and reduce mistakes.
- **New Jobs:** AI creates new job opportunities, such as AI developers and data scientists.
- **Changes in Work:** AI changes how people perform their daily tasks.
- **New Skills:** People need to learn new skills to work effectively with AI technology.
- **Job Losses:** Some people may lose their jobs when AI replaces the tasks they perform.

## Data and AI

Data is very important for AI because AI learns from data, just as students learn from books. AI uses accurate data to find patterns, make better decisions, and improve its results.

**Example:** YouTube uses data about the videos people watch to recommend similar videos they may like.

## Automatic Data Collection

Automatic data collection is the process of collecting data using computers and devices without entering it manually. AI systems use this data to learn and improve their results.

**Example:** A fitness tracker automatically collects data about steps, heart rate, and daily activity.

## Data Quality and Quantity

Data quality and data quantity are important for AI systems because they affect how well an AI system learns and produces results.

### 1. Data Quality

**Data quality** means how correct, complete, and reliable the data is. High-quality data helps AI produce better results.

**Example:** Correct student marks help AI predict student performance more accurately.

### 2. Data Quantity

**Data quantity** means the amount of data available for learning. A sufficient amount of data helps AI identify patterns more effectively.

**Example:** An AI system trained on thousands of animal pictures can learn to recognize different animals.

## Real-World Applications of AI

### 1. Natural Language Processing (NLP)

Natural Language Processing (NLP) helps computers understand and respond to human language.

**Example:** Chatbots can answer questions, and translation apps can translate languages.

### 2. Robotics in Everyday Life

AI-powered robots help people perform different tasks automatically and efficiently.

- **Robot Vacuum Cleaners:** Clean floors automatically.
- **Robotic Lawn Mowers:** Cut grass automatically.
- **Industrial Robots:** Assemble products in factories.
- **Delivery Robots:** Deliver food and packages.
- **Medical Robots:** Assist doctors during surgeries.
- **Agricultural Robots:** Help plant seeds and harvest crops.

### 3. Speech Technology and Voice Control

AI allows computers and smart devices to recognize, understand, and respond to human speech. It enables people to control devices and perform tasks using voice commands.

**Example:** Google Assistant can set alarms, play music, make calls, and answer questions using voice commands.

### 4. Recommendation Systems

AI suggests videos, products, and music based on users' interests, preferences, and past activities.

**Examples:**

- YouTube recommends videos based on what people watch.
- Netflix suggests movies and shows users may enjoy.
- Online shopping websites recommend products based on previous searches and purchases.

### 5. Autonomous Vehicles (Self-Driving Vehicles)

AI enables vehicles to understand their surroundings, navigate roads, and make driving decisions with little or no human control.

**Example:** Self-driving cars use cameras, sensors, and AI to detect traffic signs, other vehicles, and pedestrians. They use this information to make driving decisions and travel toward their destinations.

## Benefits of AI Applications

- **Saves Time:** AI completes tasks quickly, allowing people to focus on more important work.
- **Improves Education:** AI-powered learning apps help students learn according to their abilities and needs.

- **Supports Businesses:** AI helps businesses plan activities, predict sales, and provide better customer service.
- **Reduces Human Errors:** AI can perform tasks accurately when it is properly designed and trained.
- **Improves Different Fields:** AI improves work in agriculture, banking, healthcare, and transportation.

## Limitations and Challenges of AI Applications

- **Incorrect Results:** AI may produce incorrect results if it uses low-quality or biased data.
- **Misunderstanding Questions:** AI chatbots may misunderstand questions when the language is unclear.
- **Job Losses:** AI may replace workers in some jobs, such as data entry and customer service.
- **Biased Decisions:** AI may make unfair decisions if it is trained on biased data or designed poorly.
- **Need for Careful Testing:** AI systems must be tested carefully because mistakes in fields like healthcare and law can cause serious problems.

## Ethics, Impacts, and Social Impact of AI

Artificial Intelligence (AI) is changing how people live, work, and communicate. However, AI must be developed and used responsibly to protect people's rights, privacy, and safety.

### 1. Ethics in AI

Ethics in AI means using AI in a responsible, fair, and safe way. AI systems should respect human rights, protect personal information, and avoid harming people.
**Example:** An AI system should treat all students fairly when evaluating their performance.

### 2. Biases in AI and Algorithms

Bias in AI occurs when an AI system produces unfair results because of biased or unbalanced training data or a poorly designed algorithm.
**Example:** An AI hiring system may unfairly reject qualified candidates if it learns from biased hiring data.

### 3. Fairness in AI

Fairness in AI means treating people equally and avoiding unfair discrimination based on gender, age, background, or other personal characteristics.
**Example:** An AI hiring system should evaluate candidates according to their skills and qualifications.

### 4. Transparency in AI

Transparency in AI means making it clear how an AI system works and how it reaches its decisions. People should be able to understand why an AI system gives a particular result.
**Example:** A bank should explain why its AI system rejects a loan application.

### 5. Privacy

Privacy in AI means protecting people's personal information and using it only for appropriate purposes. AI systems should not collect or use personal data without proper permission.
**Example:** A learning app should protect students' names, marks, and personal details.

### 6. Data Security

Data security means protecting data from unauthorized access, theft, loss, or misuse. AI systems should use proper security measures to keep information safe.
**Example:** A hospital should protect patients' medical records from hackers.

### 7. Surveillance

Surveillance means monitoring people or their activities using cameras, sensors, or other technologies. AI can help improve security, but excessive monitoring may affect people's privacy.
**Example:** AI-powered cameras can identify suspicious activities in public places, but their use should respect people's privacy.

### 8. AI and Social Equity

AI and social equity means ensuring that the benefits of AI are available to people from different backgrounds and economic conditions. Everyone should have a fair opportunity to benefit from AI technology.
**Example:** Providing affordable AI learning tools to students in rural and urban areas can help reduce educational inequalities.

## Responsibility of AI Designers

AI designers are responsible for making AI systems fair, safe, and helpful. They must ensure that their systems work correctly and do not treat people unfairly. Creating trustworthy AI is both a technical task and a moral responsibility.

### Following Ethics in AI

AI ethics means following the principles of fairness, honesty, and responsibility when creating or using AI. Designers must protect privacy, prevent harm, and avoid discrimination. Ethical AI builds trust between people and technology.

## Exercise: Short Questions and Answers

### 1. What is Artificial Intelligence (AI)?

Artificial Intelligence (AI) is a technology that enables computers and machines to perform tasks that normally require human intelligence, such as learning, problem-solving, and decision-making.

### 2. How is AI used in daily life?

AI is used in daily life through voice assistants, recommendation systems, smart cameras, online shopping, and self-driving vehicles. It helps people complete tasks more quickly and easily.

### 3. Why does AI need large and good-quality data?

AI needs sufficient and good-quality data to learn patterns, make better decisions, and produce accurate results. Poor-quality data may lead to incorrect or biased results.

### 4. What is the role of Natural Language Processing (NLP)?

Natural Language Processing (NLP) helps computers understand, interpret, and respond to human language.
**Example:** Chatbots use NLP to understand questions and provide answers.

### 5. Give one example of robotics in everyday life.

A robot vacuum cleaner is an example of robotics in everyday life. It uses sensors and automated functions to clean floors with little human effort.

### 6. What is the purpose of speech recognition technology?

Speech recognition technology enables computers and devices to recognize spoken words and convert them into text or commands.

**Example:** People can use voice commands to set alarms or search for information.

### 7. How do recommendation systems work?

Recommendation systems analyze users' interests, preferences, and previous activities to suggest relevant content.
**Example:** YouTube recommends videos based on the videos a user watches.

### 8. What is meant by bias in AI systems?

Bias in AI occurs when an AI system produces unfair results because of biased training data or a poorly designed algorithm. It may cause the system to treat some people unfairly.

### 9. Why should AI systems be transparent and fair?

AI systems should be transparent so people can understand how they make decisions. They should also be fair to ensure that people are treated equally without discrimination based on personal characteristics or background.

**1. What is Artificial Intelligence (AI)?**

a. A technology that enables machines to perform tasks requiring human intelligence
b. A method of repairing computer hardware
c. A type of computer storage
d. A programming language

> **Answer: a. A technology that enables machines to perform tasks requiring human intelligence**

**2. What does AI stand for?**

a. Automated Internet
b. Artificial Intelligence
c. Advanced Information
d. Automatic Integration

> **Answer: b. Artificial Intelligence**

**3. What does ML stand for?**